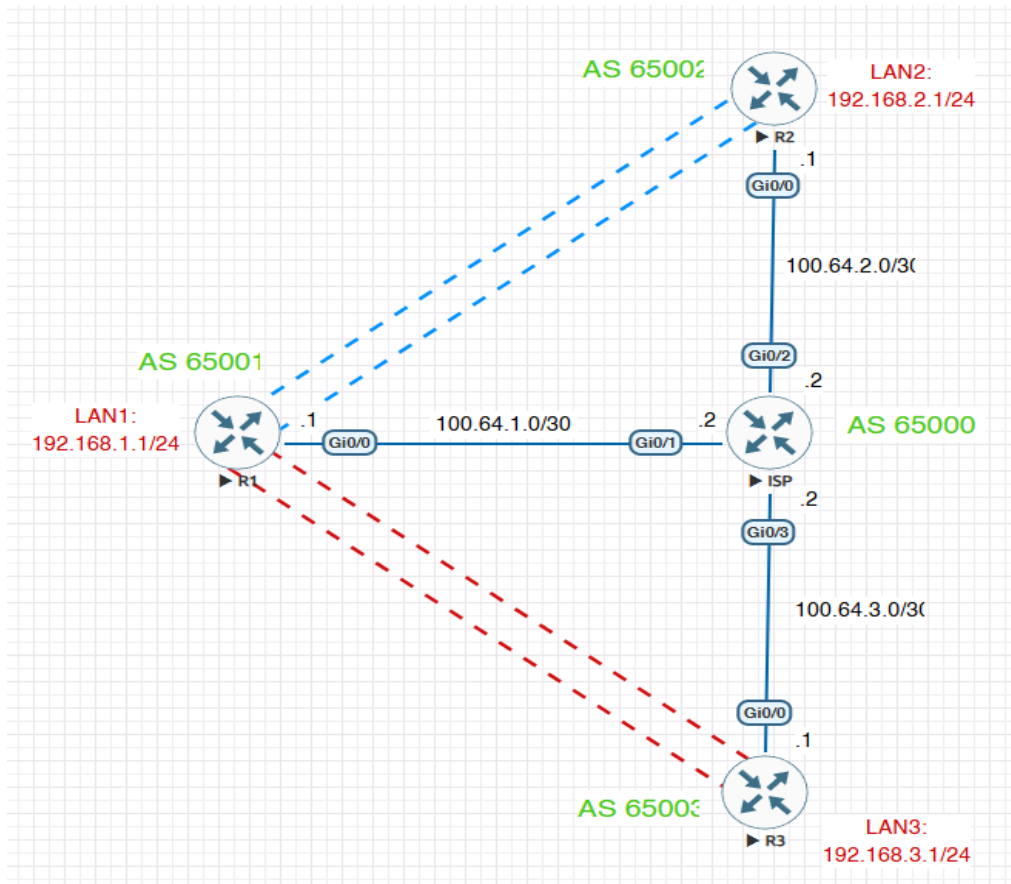


Implementation Guide: Secure Multi-Site Network Design (BGP & IPsec Analysis)

Goal: Transition from basic connectivity to architecting secure, protocol-aware infrastructure. This guide focuses on the "Why" behind the commands and the mechanics of data encapsulation.



1. The Scenario: The Untrusted Cloud

In this scenario, a regional organization needs to connect three separate office locations (R1, R2, R3) across a public Internet Service Provider (ISP). Because the ISP's network is "untrusted," we cannot simply route our private data across it in plaintext. We must build a secure "Overlay" on top of the "Underlay."

Topology and IP Schema

Device	Interface	IP Address	BGP AS	LAN Subnet	Role
ISP	Gi0/1, 0/2, 0/3	100.64.X.2/30	65000	N/A	Service Provider
R1	Gi0/0 (WAN)	100.64.1.1/30	65001	192.168.1.0/24	Branch 1 (Hub)
R2	Gi0/0 (WAN)	100.64.2.1/30	65002	192.168.2.0/24	Branch 2 (AH Peer)
R3	Gi0/0 (WAN)	100.64.3.1/30	65003	192.168.3.0/24	Branch 3 (ESP Peer)

2. Core Concepts: Understanding IPsec

IPsec (Internet Protocol Security) is a suite of protocols used to secure IP communications. It provides three main pillars:

- **Confidentiality:** Encrypting data so only the intended recipient can read it.
- **Integrity:** Ensuring the data wasn't tampered with during transit.
- **Authentication:** Verifying that the sender is who they claim to be.

AH (Authentication Header) - Protocol 51

AH provides **Integrity** and **Authentication**, but **NO Encryption**. It signs the entire packet (including the IP header). In Wireshark, you can see the data inside, but the receiver knows if even a single bit was changed.

ESP (Encapsulating Security Payload) - Protocol 50

ESP is the industry standard. It provides **Encryption**, Integrity, and Authentication. In Wireshark, everything after the ESP header is scrambled (ciphertext).

3. Step-by-Step Implementation & "The Why"

Phase 1: The BGP Underlay

Before we can secure the traffic, the routers must be able to reach each other. We use BGP because it is the language of the Internet and allows for granular control over how routes are advertised.

```
router bgp 65001
 neighbor 100.64.1.2 remote-as 65000
```

The Why: The neighbor command establishes a TCP session (Port 179) with the ISP. Without this "Underlay," the routers are isolated and the VPN cannot form.

Phase 2: ISAKMP (Phase 1 - The Handshake)

ISAKMP (IKE) is the "negotiation" phase. It is where the two routers agree on a "Secret Language" to use for their conversation. Create policy 10 on all routers.

```
crypto isakmp policy 10
  encryption aes
  hash sha
  authentication pre-share
  group 2
```

Create key to authenticate for both R2 and R3

```
crypto isakmp key CISCO address 100.64.2.1
```

The Why:

- **Encryption AES:** Tells the routers to use the AES algorithm for the handshake.
- **Hash SHA:** Ensures the handshake messages aren't tampered with.
- **Group 2:** Defines the Diffie-Hellman strength (the mathematical method for sharing keys over an insecure link).

Phase 3: Transform Sets (Phase 2 - The Data Plane)

This defines how the **actual user data** (the packets from your LAN) will be protected.

```
crypto ipsec transform-set TSET-AH ah-sha-hmac
crypto ipsec transform-set TSET-ESP esp-aes esp-sha-hmac
```

The Why: This is where you choose your "Security Flavor."

- For R1-to-R2, we use ah-sha-hmac to see the data in Wireshark while still authenticating the header.
- For R1-to-R3, we use esp-aes to fully hide the data from the ISP.

Phase 4: Crypto Access Lists (Interesting Traffic)

```
access-list 101 permit ip 192.168.10.0 0.0.0.255 192.168.20.0
0.0.0.255
```

The Why: Routers don't encrypt everything. The ACL tells the router exactly which packets are "Interesting" enough to trigger the VPN. If traffic doesn't match this list, it is sent in plaintext or dropped.

Phase 5: The Crypto Map (The Glue)

```
crypto map MYMAP 10 ipsec-isakmp
  set peer 100.64.2.1
  set transform-set TSET-AH
  match address 101
```

The Why: The Crypto Map is the "Manager." it ties the Peer (Who), the Transform Set (How), and the ACL (What) together into a single policy. **#Dont forget to assign Crypto map to interface**

4. Validation: How a Specialist Verifies

To confirm your architecture is sound, do not just rely on pings. Use these verification steps:

- **Check Phase 1:** show crypto isakmp sa. Look for QM_IDLE. This means the handshake is finished and stable.
- **Check Phase 2:** show crypto ipsec sa. Look for the "#pkts encaps" and "#pkts decaps" counters. If these are incrementing, your data is successfully entering the tunnel.
- **Verify Encapsulation:** Capture on the WAN interface. Look for **Protocol 51 (AH)** and **Protocol 50 (ESP)**.
- **Connectivity:** Ping the lan subnet of each router using the ping <ip add> source loopback 0
- **Capture Traffic :** verify the theory for AH and ESP

Running Config

#R1

```
crypto isakmp policy 1
  encr aes
  authentication pre-share
  group 2
crypto isakmp key CISCO address 100.64.2.1
crypto isakmp key CISCO address 100.64.3.1
!
!
crypto ipsec transform-set R2-AH ah-sha-hmac
  mode tunnel
crypto ipsec transform-set R3-ESP esp-aes esp-sha-hmac
  mode tunnel
!
!
crypto map MYMAP 10 ipsec-isakmp
  set peer 100.64.2.1
  set transform-set R2-AH
  match address TO-R2
crypto map MYMAP 20 ipsec-isakmp
  set peer 100.64.3.1
  set transform-set R3-ESP
  match address TO-R3
!
!
interface Loopback0
  ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet0/0
  ip address 100.64.1.1 255.255.255.252
  duplex full
  speed 1000
  media-type rj45
  crypto map MYMAP
!
!
router bgp 65001
  bgp log-neighbor-changes
  neighbor 100.64.1.2 remote-as 65000
!
!
ip access-list extended TO-R2
  permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
ip access-list extended TO-R3
  permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

#R2

```
crypto isakmp policy 1
  encr aes
  authentication pre-share
  group 2
crypto isakmp key CISCO address 100.64.1.1
!
!
crypto ipsec transform-set R2-AH ah-sha-hmac
  mode tunnel
!
!
crypto map MYMAP 10 ipsec-isakmp
  set peer 100.64.1.1
  set transform-set R2-AH
  match address TO-R1
!
!
interface Loopback0
  ip address 192.168.2.1 255.255.255.0
!
interface GigabitEthernet0/0
  ip address 100.64.2.1 255.255.255.252
  duplex full
  speed 1000
  media-type rj45
  crypto map MYMAP
!
!
router bgp 65002
  bgp log-neighbor-changes
  neighbor 100.64.2.2 remote-as 65000
!
!
ip access-list extended TO-R1
  permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
!
!
```

#R3

```
crypto isakmp policy 1
  encr aes
  authentication pre-share
  group 2
crypto isakmp key CISCO address 100.64.1.1
!
!
crypto ipsec transform-set R3-ESP esp-aes esp-sha-hmac
  mode tunnel
!
!
crypto map MYMAP 10 ipsec-isakmp
  set peer 100.64.1.1
  set transform-set R3-ESP
  match address TO-R1
!
!
interface Loopback0
  ip address 192.168.3.1 255.255.255.0
!
interface GigabitEthernet0/0
  ip address 100.64.3.1 255.255.255.252
  duplex full
  speed 1000
  media-type rj45
  crypto map MYMAP
!
!
router bgp 65003
  bgp log-neighbor-changes
  neighbor 100.64.3.2 remote-as 65000
!
!
ip access-list extended TO-R1
  permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
```

ESP

22	16.955639	100.64.3.1	100.64.1.1	ESP	182 ESP (SPI=0xea81b832)
23	16.959220	100.64.1.1	100.64.3.1	ESP	182 ESP (SPI=0x2046ba97)
24	16.963385	100.64.3.1	100.64.1.1	ESP	182 ESP (SPI=0xea81b832)

AH

4	10.594464	192.168.1.1	192.168.2.1	ICMP	158 Echo (ping) request
5	10.598472	192.168.2.1	192.168.1.1	ICMP	158 Echo (ping) reply
6	10.602064	192.168.1.1	192.168.2.1	ICMP	158 Echo (ping) request